1.1 Points, Lines, and Planes

Name _____ Period ____ Date _____

Geometry 1.1 · Student sheet

I can…

- name points, lines, and planes, and tell when points are collinear or coplanar.
- name segments, rays, and opposite rays.
- describe the intersection of lines and planes.

Vocabulary · fill in the missing word

Collinear points — _____

Coplanar points — points that lie in the same _____

Segment — part of a line that consists of two _____ and all the points between them

Ray — part of a line that _____ of an endpoint and all the points on one side of it

Opposite rays — two rays that share an _____ and form a line

Intersection — the set of points that two or more _____ have in common

Part 1 · Discovery: geometry in the classroom

Find an example in the classroom of each one.

Idea	Example in the room
A point	
A line	
A plane	
Two planes that meet	

With pencils and paper: two lines meet in at most _____ point; two planes meet in a _____.

Part 2 · Notes and examples (fill in each note, then work the example)

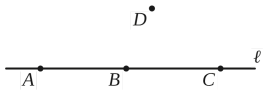
Undefined terms: a point has no size; a line extends in _____ directions with no end; a plane is flat and extends in all directions.

Example 1

a. Give two other names for line ℓ.

b. Name three collinear points.

c. Name a point that is not collinear with A and B.



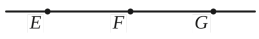
Naming segments and rays: $\overline{AB}$ is the segment; ray AB starts at _____ and goes through B forever.

Example 2

Points E, F, and G lie on a line in that order.

a. Name two opposite rays.

b. Are ray EF and ray EG the same ray?



Example 3

a. Is $\overline{AB}$ the same as $\overline{BA}$?

b. Is ray AB the same as ray BA?

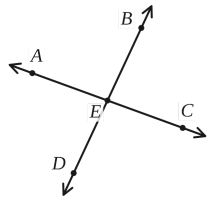
Intersections: two lines intersect in a _____; two planes intersect in a _____.

Example 4

Use the diagram.

a. What is the intersection of line AC and line BD?

b. What is the intersection of two walls of the classroom?

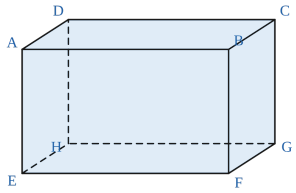


Example 5

Use the rectangular box.

a. Name the intersection of plane ABFE and plane BCGF.

b. Name three points that are coplanar with A, B, and C.



Part 3 · Partner practice

Partner 1

Points J, K, and L lie on a line in that order. Name two opposite rays and one segment.

Partner 2

In the box from Example 5, name the intersection of plane ADHE and plane EFGH.